

Personalized federated learning algorithm based on composite feature extractor

Changsheng Liu, Nanrun Zhou, Wenjun Yu*

Abstract—In the Internet of Things (IoT) environment, to maintain the global consistency and generalization capability of federated learning (FL) models with good data privacy, a personalized FL algorithm is proposed based on a composite feature extractor (CFE), i.e., FedCFE. The FedCFE framework consists of three core modules: a CFE, a conditional generator, and a knowledge distillation mechanism. By modularly combining global with local feature extractors, the proposed CFE enables device-specific personalization without disrupting global knowledge sharing. Meanwhile, the representative pseudo-data is produced by a conditional generator, which facilitates effective knowledge distillation from client models to the server model without requiring direct access to private data. Extensive experimental results demonstrate that the FedCFE consistently achieves competitive or superior performance across various heterogeneous data scenarios. On the highly non-IID CIFAR-100 dataset, the FedCFE improves classification accuracy by 2.19% compared with the state-of-the-art algorithms. These results indicate that the FedCFE provides an effective, lightweight, and privacy-preserving solution to realizing collaborative intelligence in resource-constrained and communication-unstable IoT environments.

Index Terms—Heterogeneous IoT data, Personalized federated learning, Conditional generator, Knowledge distillation, Information security.

I. INTRODUCTION

WITH the rapid development of artificial intelligence and Internet of Things (IoT), the generation and storage of data are increasingly characterized by decentralization. Traditional centralized machine learning methods are insufficient to face challenges such as data privacy protection, storage costs, and model generalization capabilities. Federated learning (FL) is an emerging distributed machine learning paradigm that trains models while protecting the privacies of participants effectively. In FL network, multiple participants contributing to a shared global model with their local data without sharing them externally aims at training the global model with

distributed data on the premise of protecting participants' privacies [1]. However, the decentralized training process and reliance on unverified local updates bring about emerging security vulnerabilities [2], [3]. In addition, in practical applications of FL, the data distributions across different clients vary significantly, making it difficult for the global model to adapt to the local data of all clients, thereby affecting or decreasing the model's accuracy and generalization capability [4], [5]. This challenging data heterogeneity issue seriously reduces the training efficiency of FL. To address this issue, the concept of personalized federated learning (PFL) was proposed. As a new branch of FL, the PFL aims to provide personalized models for each client to meet different data distributions and task requirements. PFL is suitable for the applications with significant heterogeneous data and extremely high demand on privacy protection, including face recognition [6] and medical health [7], [8]. The PFL mainly customizes personalized models via the methods, such as model decoupling, regularization constraints or meta-learning personalization [9]. These methods have demonstrated their effectiveness in different application scenarios, however they also face challenges such as increased model complexity, low training benefits, overfitting risks, high communication overhead, etc. Therefore, how to better balance personalization and global consistency to obtain greater training benefits remains an important research focus in the field of PFL. Based on the above discussions, a new PFL algorithm, i.e., FedCFE, is proposed based on a composite feature extractor (CFE). The FedCFE algorithm can further alleviate the contradiction between personalization and global consistency by integrating knowledge distillation [10] with conditional generator (CG) [11]. The main contributions of this paper are as follows:

(1) By constructing a CFE with a modular combination design approach, an innovative FedCFE algorithm is presented to effectively reduce the risk of damage or loss of private features in the local data of clients and significantly enhance the model's performance in various tasks.

(2) By integrating CG with knowledge distillation, the FedCFE realizes the data consistency between teacher model and student model. This method not only can protect the privacies of the clients, but also can bolster the performance of the student model in the heterogeneous data scenario.

(3) The personalized features are precisely extracted directly from raw data by a personalized feature extractor in the CFE module, which prevents personalized features from being damaged by the shared feature extractor,

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therefore preserving the personalized features in clients' local data further.

The structure of this paper is as follows. Section II summarizes the previously proposed PFL algorithms and analyzes their advantages and limitations. Section III mainly details an implementation process of the FedCFE. Section IV provides extensive experimental evaluations of FedCFE, encompassing ablation studies, comparison, applicability and scalability experiments, and a detailed evaluation on efficiency and system-level costs. Section V reaches a brief conclusion and discusses future research directions in the heterogeneous IoT environments.

II. RELATED WORK

The classical FL algorithm FedAvg usually obtains global model by aggregating the local models of all clients [12]. However, the data heterogeneity induces the client drift problem, which in turn deteriorates the performance of the global model [13]. The PFL method was proposed to maintain the generalization ability of the global model while adapting to the personalized needs of different clients [14]. The PFL can be divided into three categories: meta-learning personalization, regularization constraint, and model decoupling.

Meta-learning personalization methods primarily address the heterogeneity of label distributions and sample quantities. They enable each client's model to fit the characteristics of local data more accurately by introducing personalized parameters or optimization strategies. For example, the Per-FedAvg algorithm drew on the idea of meta-learning and combined FL with the model-agnostic meta-learning method, where an initial model was obtained with the data from all clients, while the final model was built after each client had performed gradient descent on the initial model locally [15]. With a meta-learning framework, the APFL algorithm determined the personalized needs of each client and then adaptively adjusted the update strength and direction of the client model according to these needs [16]. The local data distribution can be quickly fitted by the model using these methods, and consistency with the global model is maintained, thus enabling personalization and global generalization to be balanced effectively. However, when there is a large difference in clients' data distributions, highly personalized updates can cause the client model overfit the local data, thereby deviating from the goal of the global model.

Unlike the above meta-learning-based methods, regularization-based algorithms aim to mitigate excessive deviations of local model updates from the global model, which can occur due to small data volumes or uneven label distributions on clients. These algorithms align local model parameters with the global model using techniques such as L2 regularization. For example, the Ditto algorithm achieved a balance between personalization and global consistency by introducing a regularization term for each client to constrict the scope of personalized updates [17]. The FedProx algorithm added a regularization term to the local optimization objective, which is a quadratic

penalty term related to the global model parameters [18]. This regularization technique helps to tailor the amplitude of client model updates, preventing local models from deviating too far from the global model and achieving a balance between personalization and global consistency. However, improper setting of the penalty term coefficient can seriously degrade model performance. The pFedMe algorithm effectively dealt with the data heterogeneity problem in FL by treating the Moreau Envelope as a regularization loss function [19]. Although these regularization strategies do help alleviate client drift, it is difficult for these regularization strategies to balance personalization and global consistency perfectly. Since the data distributions of different clients vary apparently, the ideal degree of regularization for each client should be different. That's to say, personalized adjustments are necessary for every client. If data distributions of the clients are significantly heterogeneous, then this uniform regularization term will hinder the local model's adequate learning of local data.

Model decoupling algorithms primarily address the issues of feature heterogeneity and input distribution heterogeneity. With model decoupling, the lower layers of the neural network learn general representations of the data, which are typically required by all clients, while the higher layers capture client-specific features, enhancing the model's adaptability to local data [20]. Model decoupling algorithms can further alleviate the difficulty in balancing personalization and global consistency. For example, the FedRep algorithm divides the model into a global representation layer and a personalized classification layer, where the former layer is aggregated on the server while the latter is updated locally, thus better accommodating the data distributions of different clients [21]. The FedPer algorithm achieves model decoupling by considering the last layer of the neural network as the personalized head and the other layers as the shared body [22]. The FedALA algorithm employs an adaptive local aggregation (ALA) module to perform element-wise weighted aggregation of the global and local models before each local training session. This design dynamically incorporates global knowledge while respecting local training objectives, leading to personalized model initialization [23]. The FedBN algorithm achieves PFL by allowing each client to maintain its own batch normalization (BN) parameters without participating in global aggregation during local training. This strategy enables clients to adapt BN parameters to local data distributions, thereby improving the model's adaptability to inter-client data heterogeneity [24]. The FedAP algorithm can dynamically adjust the degree of personalization for each client, enabling the model to adaptively update its parameters according to the client's data distribution and task requirements, thus achieving personalized learning [25]. In complex FL scenarios, model decoupling algorithms can effectively alleviate the difficulty in balancing personalization and global consistency. However, in the non-IID situations, some clients have limited data due to data collection abilities or specific

scenarios. In such cases, the model is prone to overfitting by excessively memorizing noise and details in the data, which severely restricts the generalization performance of the global model. Moreover, the core design philosophy of model decoupling algorithms is to split the client's local model into body and head layers. The body layer is designed to extract general features from the data, capturing the common patterns across different clients' data. However, this design goal inherently conflicts with the need for personalization, as personalized features are the unique key information specific to a particular client's data. Therefore, compared with the original data, the personalized feature information of the data processed by the body layer will be significantly diluted. This makes subsequent personalization layers difficult to capture important personalization features.

Based on the above discussions, a new PFL algorithm called FedCFE is proposed to fundamentally solve the problems of personalized feature loss and overfitting faced by the model decoupling algorithms. First, by introducing the knowledge distillation method, the FedCFE can further alleviate the dilemma of the global model deviating from the optimal solution due to overfitting of the local models [26]. Specifically, the soft labels provided by the client teacher model enable the server student model to learn a smoother target distribution and more general feature representations, showing significant advantages in generalization ability [27]. Second, the introduction of knowledge distillation also reveals an inescapable issue, that is, the input data used by the server should be consistent with the input data used by the client teacher model when generating soft labels. However, one of the core goals of FL is to protect the data privacy of clients, that is, FL does not allow the clients to upload their raw data to the server directly [28]. This data privacy protection requirement forms a sharp contradiction with the data consistency requirement of knowledge distillation. Therefore, in the FedCFE, the CG model equipped in the server can generate image data to match the client's label data in semantics and structure. This approach not only solves the data consistency problem between the teacher and the student models in knowledge distillation, but also preserves the data privacies of the clients. Furthermore, the data generated by the CG model in the FedCFE also has rich diversity, which helps the model learn more accurate and comprehensive data distribution characteristics. Finally, to address the personalized feature loss in the model decoupling algorithms, the FedCFE integrates the feature extractors of the global and the local models by a modular combination strategy. Specifically, a composite feature extraction module is constructed by connecting the body layers of the global and the local models in parallel. This modular design ensures the client model to better capture the general features and the personalized ones in local data as much as possible, effectively solving the problem of personalized feature loss in the model decoupling algorithms.

III. METHOD

A. Formulation of FedCFE

A client-side personalized model w_{CFE} will be constructed by a modular combination approach, which consists of two feature extractors and a classifier, all of which are parameterized functions. The shared feature extractor $\bar{w}_s : \mathbb{R}^h \rightarrow \mathbb{R}^a$ and the personalized feature extractor $\bar{w}_p : \mathbb{R}^h \rightarrow \mathbb{R}^b$ map the high-dimensional local data of the client into the low-dimensional common features and personalized features, respectively. The classifier $h_p : \mathbb{R}^{a+b} \rightarrow \mathbb{R}^y$ concatenates the feature vectors extracted from the two feature extractors and then transforms them into the label dimension. The major notations used throughout this section are collected in **Appendix**. The structural schematic of the CFE model is shown in Fig. 1. Therefore,

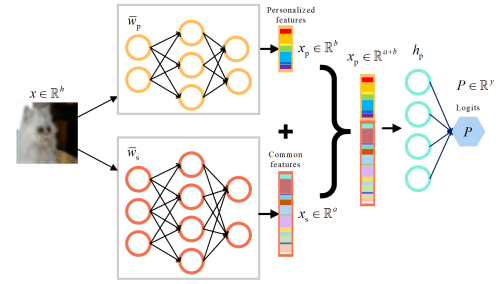


Fig. 1: Structural schematic of CFE model.

the loss function of the client can be defined as

$$L(w_{\text{CFE},k}) = \mathbb{E}_{(x_i, y_i) \sim D_k} [\ell(\bar{w}_s, \bar{w}_{p,k}, h_{p,k}; (x_i, y_i))] \quad (1)$$

where $w_{\text{CFE},k}$ is the CFE model of client k ; $\bar{w}_s$ is the feature extractor of the server model w_s ; $\bar{w}_{p,k}$ is the feature extractor of personalized model $w_{p,k}$ of client k ; $h_{p,k}$ is the personalized classifier of the k -th client's model $w_{\text{CFE},k}$. Through local iteration, each client will obtain a local optimal solution, and a globally optimal server model w_s will be obtained after T times of globally collaborative training. Therefore, the global optimization objective of the proposed algorithm can be written as

$$w_s = \arg \min_{w_s \in W_s} \left[\frac{1}{|K|} \sum_{k=1}^K \min_{w_{\text{CFE},k} \in W_{\text{CFE}}} L(w_{\text{CFE},k}) \right] \quad (2)$$

where w_s is the server model, and $w_{\text{CFE},k}$ is the CFE model of the k -th client. The overall mechanism of the proposed FedCFE algorithm is shown in Fig. 2.

B. Proposed FedCFE algorithm

1) *Server update*: Pre-trained conditional generator (PreCG). To accelerate the convergence of client models, the server pre-trains the CG model using a generative adversarial network [29]. The server needs to collect a certain amount of dataset D_s in advance. In practical IoT systems, public datasets can be obtained from open IoT platforms or from non-sensitive data collected during the initial deployment phase. It is important to note that FedCFE does not fundamentally rely on the existence

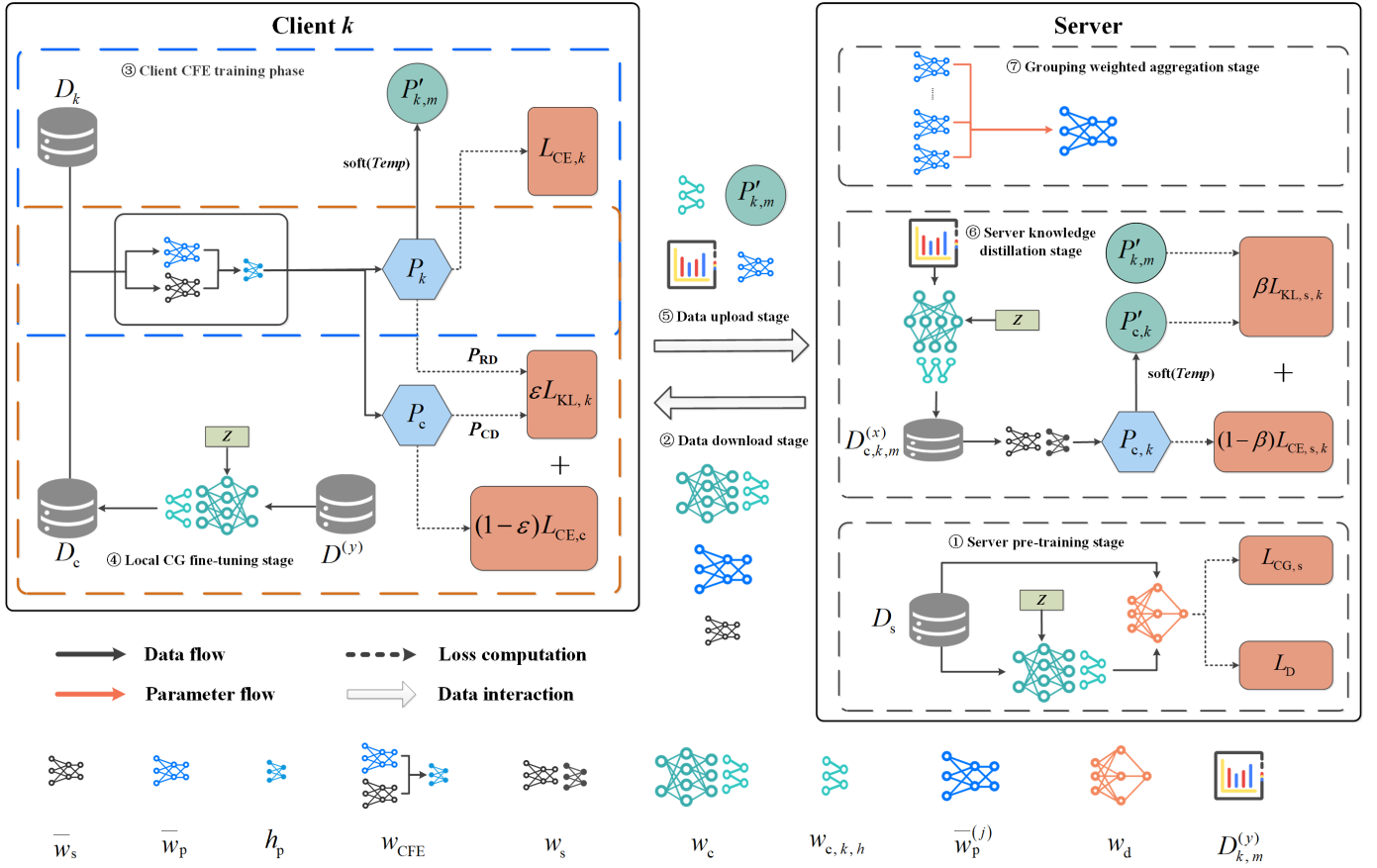


Fig. 2: Overall framework of the FedCFE algorithm.

of public datasets. In this framework, public data serve only to accelerate the initialization of the conditional generator, thereby reducing the computational burden on clients. When public data are unavailable, the CG can be trained directly on each client's local private data and uploaded to the server only during the initial communication round. Since the objective of the CG is to model the conditional data distribution of each client rather than directly optimize prediction accuracy, its parameters remain stable during subsequent training and do not require continuous fine-tuning. Under this setting, the server-side student model can perform knowledge distillation based on pseudo-data that closely match the client's real data distribution, thereby ensuring and potentially enhancing the effectiveness of distillation. Consequently, the absence of public data merely shifts part of the computational cost to the clients, without degrading convergence behavior or the final model performance.

In the experimental setup, we randomly and uniformly sample 5%-10% of the dataset as D_s . D_s is strictly disjoint with all local client datasets. This setup prevents potential bias caused by data intersection and ensures the objectivity of the experimental results as well as the independence of data partitioning. Additionally, it is necessary to initialize a CG model w_c and a discriminator w_d , which form the core components of the generative adversarial network. The CG requires a label data y

and a noise vector $\mathbf{z}$ to be combined into an embedding vector $\mathbf{E}$. In this way, discrete labels are transformed into continuous and differentiable vector representations, enabling gradient-based optimization and facilitating the learning of similarities and differences among categories [30], [31]. The introduction of the noise vector $\mathbf{z}$ enables the generator to generate different images under the same label y , thereby increasing the diversity of the generation results [32], [33]. Therefore, the cross-entropy (CE) loss function L_{CG} of the CG can be defined as

$$L_{CG,s} = -\mathbb{E}_{\mathbf{z} \sim p(\mathbf{z}), y \sim p(y)} [\ln w_d(w_c(\mathbf{z}, y), y)] \quad (3)$$

Since the goal of the discriminator is to distinguish between real samples and generated ones, the loss function of the discriminator is defined as the binary CE loss,

$$L_D = -\mathbb{E}_{x \sim p(x), y \sim p(y)} [\ln w_d(x, y)] - \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z}), y \sim p(y)} [\ln (1 - w_d(w_c(\mathbf{z}, y), y))] \quad (4)$$

These two loss functions jointly drive the CG to generate more realistic data. The server's PreCG model w_{cs} is sent to the client only during the first round of communication.

Server knowledge distillation. The soft labels in knowledge distillation contain richer inter-class relationship information than hard labels and can provide a smoother target distribution [34]–[36]. During the t -th round of communication, the label data of client k and the

CG model are used to generate the training data required by the server model.

$$D_{c,k,m}^{(x)} = w_{c,k}^{(t+1)} \left(D_{k,m}^{(y)}, \mathbf{z} \right) \quad (5)$$

where $w_{c,k}^{(t+1)}$ is the client k local CG that the server reconstructs based on the last h -layer parameters of the locally fine-tuned CG uploaded by client k ; $D_{k,m}^{(y)}$ is the statistical information of the label sub-dataset composed of $m\%$ pieces of data randomly sampled from the label dataset D_k according to the proportion of each image category of the client k ; $D_{c,k,m}^{(x)}$ is the dataset generated by the CG of the k -th client using $D_{k,m}^{(y)}$ as the conditioning information in the t -th communication round. The CE loss function $L_{CE,s,k}$ of the server model can be expressed as

$$L_{CE,s,k} = - \frac{1}{|D_{c,k,m}^{(x)}|} \sum_{x_{c,k} \in D_{c,k,m}^{(x)}} Q(x_{c,k}) \ln P(x_{c,k}) \quad (6)$$

where $|D_{c,k,m}^{(x)}|$ is the number of samples in dataset $D_{c,k,m}^{(x)}$; $P(x_{c,k})$ is the predicted probability distribution of the generated data, while $Q(x_{c,k})$ is the true probability distribution of the generated data. The Kullback-Leibler (KL) divergence loss $L_{KL,s,k}$ between the soft labels uploaded by client k and the soft labels of the data generated by $w_{c,k}^{(t+1)}$ is calculated. $L_{CE,s,k}$ and $L_{KL,s,k}$ together constitute the total loss function L_{KD} of the server student model,

$$\begin{aligned} L_{KD} &= \beta L_{KL,s,k} + (1 - \beta) L_{CE,s,k} \\ &= \frac{\beta}{K} \sum_{k=1}^K \left[\frac{1}{|D_{c,k,m}^{(x)}|} \sum_{x_{c,k} \in D_{c,k,m}^{(x)}} P'_k \ln \frac{P'_k}{P'_{c,k}(x_{c,k})} \right] - \\ &\quad \frac{1 - \beta}{K} \sum_{k=1}^K \left[\frac{1}{|D_{c,k,m}^{(x)}|} \sum_{x_{c,k} \in D_{c,k,m}^{(x)}} Q(x_{c,k}) \ln P(x_{c,k}) \right] \end{aligned} \quad (7)$$

where $P'_{c,k}(x_{c,k})$ is the soft label calculated from the generated dataset $D_{c,k,m}^{(x)}$; P'_k is the soft label uploaded by client k ; β is an interpolation weight to balance the contributions of the KL divergence loss and the CE loss, which enables the server student model to consider both the accuracy of category prediction and the similarity of the probability distribution with the teacher model during the training process. Therefore, the global objective is

$$w_s^{(t+1)} = \arg \min_{w_s^{(t)}} L_{KD} \left(w_s^{(t)} \right) \quad (8)$$

where $w_s^{(t)}$ is the server model during the t -th round of communication. Based on the above global loss function, the gradient descent algorithm and its variants can be used to calculate the gradient $\nabla L_{KD} \left(w_s^{(t)} \right)$ of the server model parameters.

$$w_s^{(t+1)} = w_s^{(t)} - \eta \nabla_{w_s} L_{KD} \left(w_s^{(t)} \right) \quad (9)$$

where η is the learning rate. After multiple rounds of iterative updates, when the global loss function $L_{KD} \left(w_s^{(t)} \right)$

converges to a local minimum or reaches the preset iteration termination condition, a new round of the global optimal model $w_s^{(t+1)}$ can be obtained.

Grouped weighted aggregation of personalized feature extractors. To enhance global collaboration efficiency while maintaining the representational capacity of personalized models, clients are first clustered according to their data distribution characteristics. After every 5-10 rounds of client-server collaborative training, weighted aggregation of personalized feature extractors $\bar{w}_{p,k}^{(t)}$ is performed within each cluster C_j , where the aggregation weights are determined by the size of each client's local dataset D_k . Therefore, similar statistical properties are shared among clients within the same cluster, thereby promoting effective intra-group knowledge collaboration,

$$\bar{w}_p^{(t,j)} = \sum_{k \in C_j} \frac{|D_k|}{\sum_{i \in C_j} |D_i|} \bar{w}_{p,k}^{(t)} \quad (10)$$

where $\bar{w}_{p,k}^{(t)}$ represents the personalized feature extractor of client k during the t -th round of communication; $\bar{w}_p^{(t,j)}$ denotes the aggregated personalized feature extractor corresponding to the j -th client cluster during the t -th round of communication. After aggregation, the server distributes the aggregated parameter $\bar{w}_p^{(t,j)}$ back to the corresponding clients to replace their local $\bar{w}_p$, upon which clients resume local training. Through periodic weighted aggregation, the framework preserves each client's personalized characteristics while alleviating overfitting induced by prolonged local training. This mechanism facilitates knowledge sharing among local models without compromising personalized feature representations.

2) Client update: CFE model update. When updating the w_{CFE} model of the client, the parameters of $\bar{w}_s$ need to be frozen, and only the parameters of $\bar{w}_p$ and h_p are updated. Freezing the parameters of the server's feature extractor helps maintain the stability of the model, since these parameters have been fully trained on the server and have good generalization ability and common feature extraction ability. The local client continuously iterates locally and updates the local model through gradients,

$$h_p^{(e+1)} = h_p^{(e)} - \eta \nabla_{h_p} L \left(w_{CFE}^{(e)} \right) \quad (11)$$

$$\bar{w}_p^{(e+1)} = \bar{w}_p^{(e)} - \eta \nabla_{\bar{w}_p} L \left(w_{CFE}^{(e)} \right) \quad (12)$$

where $w_{CFE}^{(e)}$ is the CFE model parameters of the e -th local iteration of the client; $h_p^{(e)}$ and $\bar{w}_p^{(e)}$ represent the CFE model parameters associated with h_p and $\bar{w}_p$ of the client at the e -th local iteration, respectively.

Local CG model update. The client initializes the local $w_{c,k}^{(t)}$ with w_{cs} sent by the server. The client k takes the label set $D_k^{(y)}$ and the noise vector $\mathbf{z}$ as the input data of $w_{c,k}^{(t)}$ to generate data $D_{c,k}^{(x)}$ corresponding to $D_k^{(y)}$. Its

CE loss L_{CE} is calculated as

$$L_{CE,c} = \mathbb{E}_{y \sim p(y)} \left[\sum_{\mathbf{z} \sim p(\mathbf{z})} Q(y) \ln P_{CD} \left(w_{c,k}^{(t)}(\mathbf{z}, y) \right) \right] \quad (13)$$

where $Q(y)$ is the real probability distribution of the client's local data, i.e., one-hot encoding; P_{CD} is the predicted distribution of the generated data. The KL divergence loss L_{KL} is calculated as

$$L_{KL,k} = \mathbb{E}_{\substack{x \sim p(x) \\ y \sim p(y) \\ \mathbf{z} \sim p(\mathbf{z})}} \left[\sum_{\substack{x \in D_k^{(x)} \\ y \in D_k^{(y)}}} P_{RD}(x) \ln \frac{P_{RD}(x)}{P_{CD} \left(w_{c,k}^{(t)}(\mathbf{z}, y) \right)} \right] \quad (14)$$

where P_{RD} is the predicted distribution of the real data. The CG loss function $L_{CG,k}$ after $L_{CE,c}$ and $L_{KL,c}$ weighted average processing is defined as

$$L_{CG,k} = \varepsilon L_{KL,c} (P_{RD} \| P_{CD}) + (1 - \varepsilon) L_{CE,c} \left(\mathbf{z}, D_k^{(y)} \right) \quad (15)$$

This loss function is used to locally update the last h layer parameters of $w_{c,k}^{(t)}$, and other layer parameters are frozen. The goal of the client's local CG is

$$w_{c,k}^{(t+1)} = w_{c,k}^{(t,e+1)} = \arg \min_{w_{c,k}^{(t,e)}} L_{CG} \left((w_{c,l}; w_{c,k}^{(t,e)}, w_{CFE,k}^{(t,e)}, D_k^{(y)}, \mathbf{z}) \right) \quad (16)$$

where $w_{c,k}^{(t,e)}$ represents the last h layers of the CG of the client k in round t , iteration e ; $w_{c,l}$ represents the weight parameter of the remaining low-dimensional l layers in the CG.

Finally, $m\%$ of the data $D_{k,m}$ is randomly sampled from the data of client k . To ensure that the sampled data can accurately reflect the data distribution characteristics of client k , the sampling process strictly follows the proportion of each image category of client k . Each client uploads the fine-tuned parameters of the last h layers of its $w_{c,k}$, the soft label set $P'_{k,m}$, and the statistical information $D_{k,m}^{(y)}$ of the small sampled subset. Note that these uploaded contents are abstract model-level or statistical representations rather than raw IoT data, thus largely preventing the leakage of individual sensor readings or data. The whole process is shown in Algorithm 1.

C. Privacy and Security Discussion

In typical IoT environments, the massive number of devices, limited network bandwidth, and unstable communication impose stringent requirements on the security and efficiency of federated learning frameworks [37], [38]. The FedCFE framework addresses these challenges through communication and privacy optimization. During the collaborative training between clients and the server, only processed high-level parameters and statistical information are exchanged. Specifically, clients upload the fine-tuned parameters of the generator's h layers, a small set of soft labels, the label distribution information of

sampled subsets, and periodically, the personalized feature extractor. The server then distributes the updated shared feature extractor module and the aggregated personalized feature extractors. This lightweight communication mechanism significantly reduces communication overhead and energy consumption, avoiding the direct transmission of raw samples and full gradients, thereby enhancing privacy protection and system stability in IoT network environments.

However, since IoT devices are typically deployed in open and heterogeneous edge networks, the FedCFE communication mechanism may still face multiple security threats. First, model inversion and membership inference attacks may exploit uploaded high-level parameters or soft labels to infer partial data characteristics from edge devices. Second, model poisoning and backdoor attacks could arise when compromised IoT devices inject malicious parameters to disrupt the server aggregation process [39]. Moreover, under extreme non-IID conditions, label distribution statistics may still implicitly reveal partial data distribution property, introducing risks of distribution inference.

To address these challenges, future work will focus on enhancing the security and robustness of our FedCFE in IoT scenarios. Differential privacy mechanisms will be introduced to inject controlled noise into parameter and soft label uploads, mitigating model inversion risks. Secure Aggregation and Homomorphic Encryption techniques will be integrated to enable the server to perform aggregation without directly accessing the raw uploaded contents. Furthermore, a model signing and integrity verification module can be deployed on the server side, combined with IoT node log tracking, to ensure traceability and tamper-resistance throughout the model update process.

IV. EXPERIMENTS

A. Experiment configuration

Dataset settings. In experiments focusing on visual classification tasks, the Fashion-MNIST, CIFAR-10, and CIFAR-100 datasets are involved, while the HAR and AGNews datasets are adopted for non-visual classification tasks. To simulate real-world client data heterogeneity, a Dirichlet-based non-IID partitioning method is adopted, where different heterogeneity levels are controlled by the parameter α . To thoroughly investigate the impact of heterogeneity on the algorithm performances, three representative α values 0.5, 0.3, 0.1 are taken to perform heterogeneous processing on the datasets. Fig. 3 presents the client data distribution on the CIFAR-10 dataset under different α values as a representative example. The same Dirichlet-based partitioning strategy is consistently applied to all other evaluated datasets, resulting in similar heterogeneity characteristics. The histograms and heatmaps demonstrate that the data types owned by the clients become more diverse as α increases, while the color contrast in the heatmaps diminishes, indicating that the sample size across categories is more uniform.

Algorithm 1 FedCFE

Input: The communication rounds T , per-round iteration numbers of the CFE (τ_i), CG (τ_c), server model (τ_s), and PreCG (τ_{pc}), the client participation rate ρ , and the aggregation interval T_p for the personalized feature extractor.

Output: Local models $w_{CFE,1}^{(T)}, w_{CFE,2}^{(T)}, \dots, w_{CFE,K}^{(T)}$.

ServerUpdate:

- 1: The server initializes w_c , w_d and w_s .
- 2: **for** $e = 1, 2, \dots, \tau_{pc}$ **do**
- 3: Update w_c and w_d according to Eqs. (3) and (4), respectively.
- 4: $w_{cs} \leftarrow w_c^{(\tau_{pc})}$.
- 5: Server sends w_{cs} to all clients to initialize $w_c^{(t)}$.
- 6: **for** $t = 1, 2, \dots, T$ **do**
- 7: Server samples client subset $\mathcal{K}^{(t)}$ according to ρ .
- 8: **for** client $k \in \mathcal{K}^{(t)}$ in parallel **do**
- 9: ClientUpdate().
- 10: Reconstruct the client's $w_{c,k}^{(t+1)}$ with $w_{c,k,h}^{(t+1)}$, w_c , l .
- 11: **for** $e = 1, 2, \dots, \tau_s$ **do**
- 12: Update the $w_s^{(t,e)}$ according to Eqs. (8) and (9).
- 13: $w_s^{(t+1)} \leftarrow w_s^{(t,\tau_s)}$.
- 14: if $t \% T_p = 0$:
- 15: The server clusters clients based on their data-distribution
 ClientClustering $\left(\left\{ D_{i,m}^{(y)} \right\}_{i \in \mathcal{K}^{(t)}} \right)$.
- 16: The server then performs aggregation according to Eq. (10) and sends the result $\bar{w}_p^{(t,j)}$ to all clients within the corresponding cluster.
- 17: **Return** $w_{CFE,1}^{(T)}, w_{CFE,2}^{(T)}, \dots, w_{CFE,K}^{(T)}$.

ClientUpdate:

- 18: **for** $e = 1, 2, \dots, \tau_i$ **do**
- 19: Freeze $\bar{w}_s^{(t)}$ in $w_{CFE}^{(t,e-1)}$ and update the client's CFE according to Eqs. (11) and (12).
- 20: **for** $e = 1, 2, \dots, \tau_c$ **do**
- 21: Freeze $w_{c,k,l}$ in $w_{c,k}^{(t,e-1)}$ and update the client's CG model according to Eqs. (15) and (16).
- 22: if $t \% T_p = 0$:
- 23: return $w_{c,k,h}^{(t+1)}$, $P'_{k,m}$, $D_{k,m}^{(y)}$, $\bar{w}_p^{(t,k)}$.
- 24: else:
- 25: return $w_{c,k,h}^{(t+1)}$, $P'_{k,m}$, $D_{k,m}^{(y)}$.

Baseline algorithms. To evaluate algorithm performance, several representative FL and PFL methods were selected for comparison. FedAvg assumes IID client data and learns a global model through collaborative training, while PFL methods are more suitable for non-IID settings [12]. Specifically, FedProx introduces a proximal term to improve robustness on non-IID data [18]. APFL adaptively balances personalization and global consistency [16]. Per-FedAvg enables personalized models via local adaptation from a shared initialization [15]. Ditto constrains personalized updates with regularization [17]. FedRep separates global and local model components to alleviate

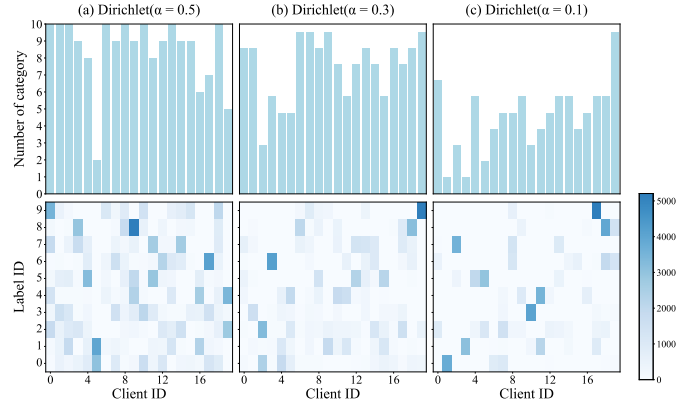


Fig. 3: Distribution of client data with three different degrees of heterogeneity ($\alpha = 0.5, 0.3, 0.1$) on the CIFAR-10 dataset.

heterogeneity [21]. FedALA mitigates data heterogeneity by incorporating an ALA module [23].

Experimental parameter settings. The global communication rounds for each algorithm is set to 50, and the initial number of local iterations is set to 5. The proposed FedCFE algorithm adaptively adjusts the number of local iterations from 1 to 5 according to the amount of client data. Except for the scalability experiments in Section IV-E, all other experiments are conducted with 20 clients. The mini-batch size is set to 32. The learning rate is uniformly set to 0.01, and the learning rate decay is set to 0.8. The learning rate decay operation is performed once every 5 rounds of global communication.

Model configuration. In the PreCG stage of the server, for the CIFAR-100 dataset, the PreCG model consists of five fully connected layers, and the corresponding discriminator consists of four convolutional layers and one fully connected layer. The network used by the server model is CNN, which contains two convolutional layers and two fully connected layers. The architecture of the CG model deployed on the client is consistent with that of the server-side CG model. It is worth noting that, across all experiments, the prediction model deployed in FedCFE contains approximately 0.75 times the number of parameters compared to the baseline methods. This lightweight design is particularly suitable for resource-constrained IoT and edge learning scenarios, enabling more efficient model deployment. At the same time, it maintains competitive predictive performance, thereby achieving a better balance between accuracy and efficiency.

B. Comparative experiment

Comparative analysis of accuracy. The performances of various FL algorithms under the combined scenarios of five different datasets and three degrees of data heterogeneity are investigated. To ensure the reliability and the stability of experimental results, each experiment is repeated 5 times to obtain the average model accuracy. Tables I and II summarize the results on visual datasets

TABLE I: Accuracy (%) comparison of different algorithms on visual datasets under different heterogeneity levels.

Dataset	Fashion-MNIST			CIFAR-10			CIFAR-100		
	$\alpha = 0.5$	$\alpha = 0.3$	$\alpha = 0.1$	$\alpha = 0.5$	$\alpha = 0.3$	$\alpha = 0.1$	$\alpha = 0.5$	$\alpha = 0.3$	$\alpha = 0.1$
FedAvg [12]	89.58±0.16	88.42±0.57	86.36±1.64	64.27±0.20	60.72±1.67	54.60±0.69	37.41±0.57	35.15±0.24	30.49±0.45
Per-FedAvg [15]	88.62±0.01	89.55±0.15	94.74±0.10	62.73±0.76	65.74±1.62	82.65±0.68	32.14±0.20	33.59±0.09	36.87±0.90
APFL [16]	92.73±0.05	93.38±0.04	96.84±0.02	72.54±0.24	77.38±0.30	88.95±0.10	36.85±0.17	44.09±0.18	57.35±0.23
FedProx [18]	89.65±0.28	88.94±0.05	87.61±0.27	64.35±0.13	60.08±0.05	55.47±1.20	38.01±0.24	36.92±0.04	30.66±0.04
Ditto [17]	91.20±0.05	93.23±0.02	97.24±0.07	70.00±0.22	78.06±0.16	88.13±0.08	31.46±0.02	39.93±0.01	54.33±0.20
FedALA [23]	92.16±0.03	93.40±0.21	97.38±0.07	72.50±0.18	77.12±0.56	88.53±0.24	38.91±0.16	41.52±0.57	57.39±0.53
FedCFE (Ours)	91.75±0.09	93.74±0.05	97.34±0.10	72.35±0.31	79.77±0.02	89.08±0.02	38.05±0.08	45.53±0.15	59.58±0.29

TABLE II: Accuracy (%) comparison of different algorithms on non-visual datasets under different heterogeneity levels.

Dataset	HAR			AGNews		
	$\alpha = 0.5$	$\alpha = 0.3$	$\alpha = 0.1$	$\alpha = 0.5$	$\alpha = 0.3$	$\alpha = 0.1$
FedAvg [12]	90.74±0.05	80.80±0.13	76.59±0.26	81.34±1.68	82.47±4.60	81.75±0.34
Per-FedAvg [15]	90.79±0.91	93.08±0.19	93.82±0.23	69.46±3.70	81.65±0.49	85.83±0.13
APFL [16]	91.19±0.18	95.02±0.02	97.11±0.05	84.92±0.12	90.89±0.28	91.87±0.12
FedProx [18]	90.16±0.02	83.35±0.30	75.29±0.43	82.95±0.58	89.34±0.02	86.91±0.47
Ditto [17]	86.90±0.02	94.14±0.20	97.35±0.09	80.35±0.05	88.29±0.23	90.28±0.27
FedALA [23]	91.33±0.48	93.41±0.63	95.78±0.40	85.14±0.85	90.21±0.98	92.06±0.57
FedCFE (Ours)	91.07±0.02	95.04±0.12	98.32±0.02	84.89±0.21	90.99±0.16	92.59±0.22

and non-visual datasets, respectively. The two best results are highlighted in bold.

It can be seen from Table I and Table II that the higher the degree of data heterogeneity in the dataset is, the higher the test accuracy of the PFL algorithm is. The test accuracy of the proposed FedCFE is either better than that of the other SOTA algorithms or close to the test accuracy of the optimal algorithm. On the CIFAR-100 dataset, the test accuracy for FedCFE under the heterogeneity degree α taking 0.1 can reach 59.51%, which is 2.12% higher than that of the suboptimal algorithm FedALA. To further validate the generalization ability of the FedCFE beyond visual benchmarks, we also evaluate it on two non-visual IoT datasets: HAR and AGNews. Under high heterogeneity ($\alpha = 0.1$), the FedCFE achieves the best performance on both datasets, with 98.32% accuracy on HAR and 92.59% on AGNews, surpassing all competing models. These results demonstrate that the FedCFE effectively generalizes to multimodal IoT data, maintaining strong personalization capability and robustness across diverse modalities. In addition, to intuitively illustrate the performance changes of the algorithms under different experimental settings, the accuracy convergence curves of the algorithms are presented in Fig. 4.

It can be intuitively observed from Fig. 4 that the accuracy convergence of the proposed FedCFE algorithm shows a more stable trend and its fluctuation range is significantly smaller than that of most baseline algorithms. This phenomenon strongly verifies that the FedCFE algorithm has excellent robustness. On datasets with higher task complexity or stronger data heterogeneity, such as CIFAR-100 and HAR, FedCFE demonstrates more pro-

nounced accuracy gains over competing methods. For some datasets with relatively lower complexity or near-saturated performance, the accuracy differences among PFL methods become less distinguishable. In these cases, FedCFE achieves performance comparable to the strongest baselines while employing a substantially more lightweight prediction model. Specifically, the prediction model deployed in FedCFE contains approximately 0.75 times the number of parameters used by other methods, which constrains model capacity but improves efficiency. This lightweight design partially explains the reduced accuracy gaps observed in certain scenarios, while still maintaining competitive performance.

C. Ablation experiment

Influence of hyperparameters in the CFE model. w_{CFE} has two main hyperparameters, i.e., m and L_p . The hyperparameter L_p is the number of layers of the personalized feature extractor $\bar{w}_p$ in w_{CFE} , which directly affects the accuracy of the model in capturing the personalized features of client data. The hyperparameter m refers to extracting $m\%$ real-label data from the client's private data and uploading it to the server.

Influence of the hyperparameter L_p . The influence of the number of layers in $\bar{w}_p$ on the performance of the CFE model is shown in Table III, where $m = 5$. Although the accuracy of the CFE model rises with the increase of the number of layers in $\bar{w}_p$, the number of parameters in the CFE model also increases. When L_p taking 2 or 3, the performance improvement in accuracy slows down. After weighing the influence of the number of layers of $\bar{w}_p$ on

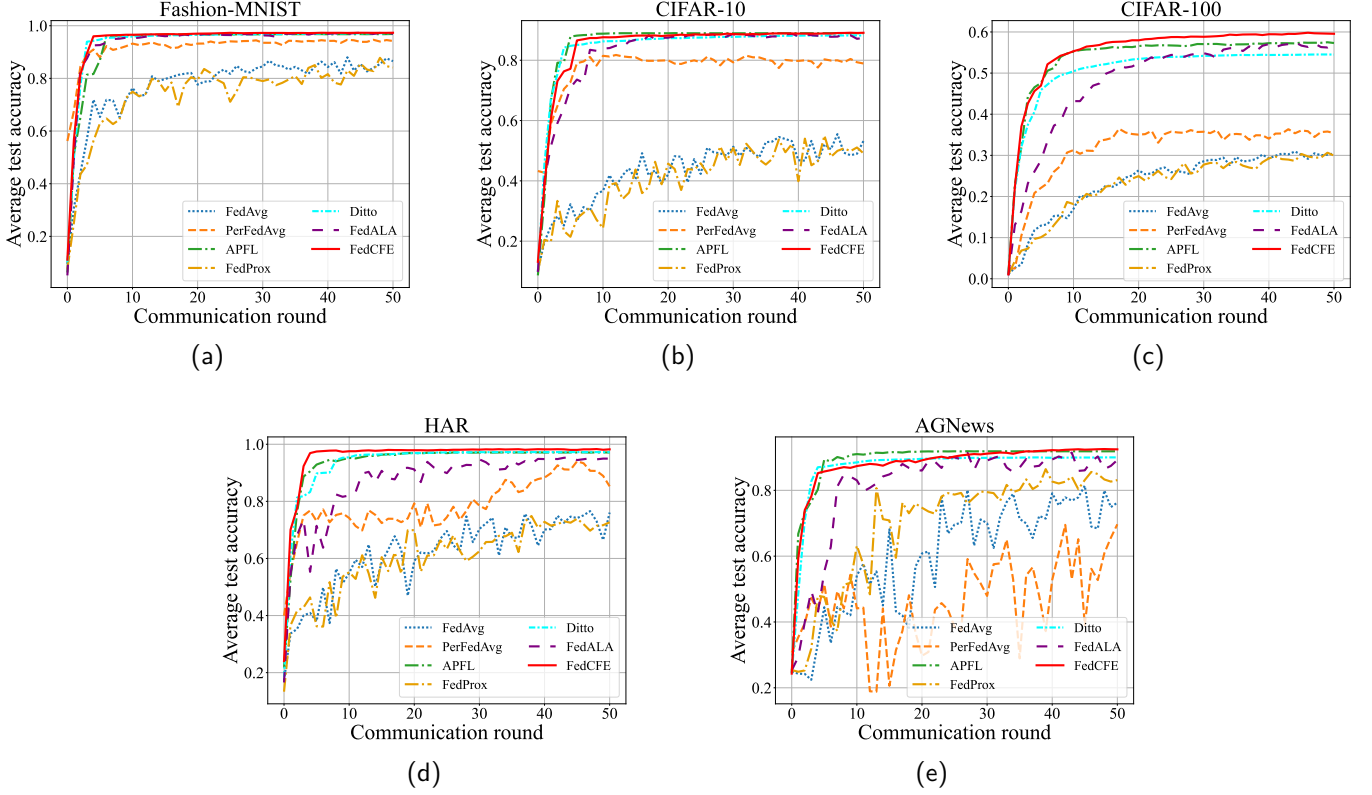


Fig. 4: The performances of different FL algorithms under the degree of heterogeneity Dirichlet(0.1) on the datasets of Fashion -MNIST, CIFAR-10, CIFAR-100, HAR and AGNews.

TABLE III: Ablation of L_p on CIFAR-10 accuracy (%).

Hyperparam.	$m = 5$			
	$L_p = 0$	$L_p = 1$	$L_p = 2$	$L_p = 3$
Acc.	81.79±0.02	86.29±0.15	88.58±0.12	89.08±0.02
Param.	260,362	261,882	410,524	438,451

TABLE IV: Ablation of m on CIFAR-10 accuracy (%).

Hyperparam.	$L_p = 2$			
	$m = 1$	$m = 5$	$m = 10$	$m = 20$
Acc.	87.26±0.12	88.58±0.12	88.71±0.04	88.79±0.07
Param.	410,524			

the accuracy and computational complexity of the CFE model, the hyperparameter L_p of the FedCFE is set to 2.

Fig. 5(a) presents the influence of different numbers of layers of $\bar{w}_p$ on the test accuracy of the CFE model. If $L_p = 0$, the number of layers of $\bar{w}_p$ in the CFE model is 0, in other words, there is no personalized feature extractor. Moreover, the test accuracy of the model with $L_p = 0$ is significantly lower than that under the condition such that $L_p = 1, 2, 3$. It is indicated that $\bar{w}_p$ can extract more useful personalized features from the data. This also indirectly verifies that the number of personalized features contained in the data usually decreases compared with the original data after being processed by the body layer. This is because the main design goal of the body layer is to extract the public features in the data to achieve knowledge sharing and fusion among different clients. During this process, some personalized detailed information is smoothed or ignored, thus the importance of the personalized feature extractor is highlighted in capturing and retaining the personalized features of the data.

Influence of hyperparameter m . With $L_p = 2$, the influence of hyperparameter m on the accuracy of the CFE model is shown in Fig. 5(b). The corresponding curve with $m = 1$ lies below the curves when m takes other values. The scale of the training data used by the server during knowledge distillation is too small. This makes the public feature extractor of the server model unable to learn relatively comprehensive knowledge, which in turn degrades the generalization ability of the CFE. As shown in Table IV, it can be found that the improvement of the accuracy of the CFE model slows down if $m = 5$ or larger.

D. Applicability experiment of FedCFE

The proposed FedCFE algorithm adds a personalized feature extractor module to the original FL model architecture, and introduces a knowledge distillation mechanism and CG to assist model training. These improvements enable FedCFE to be flexibly integrated into existing PFL algorithms, thereby providing a feasible technical path for enhancing their performance in PFL scenarios.

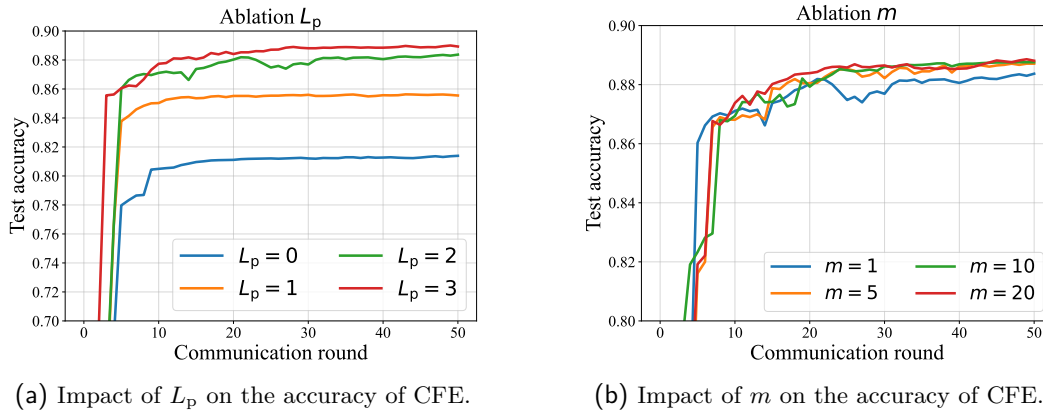


Fig. 5: Ablation study on the impact of L_p and m on the accuracy of CFE.

TABLE V: Accuracy improvements of other FL algorithms after being integrated with the FedCFE algorithm under heterogeneous data distribution Dirichlet($\alpha = 0.1$).

Dataset	CIFAR-10/CIFAR-100	
Method	Accuracy (%)	Improvement
FedAvg+CFE	89.06 $\pm$ 0.07 / 59.51 $\pm$ 0.30	34.46 / 20.02
Per-FedAvg+CFE	87.08 $\pm$ 0.09 / 52.89 $\pm$ 0.08	4.43 / 16.02
APFL+CFE	89.05 $\pm$ 0.03 / 59.48 $\pm$ 0.08	0.10 / 2.03
FedProx+CFE	83.21 $\pm$ 0.12 / 37.67 $\pm$ 0.15	27.74 / 7.01
Ditto+CFE	89.62 $\pm$ 0.04 / 60.90 $\pm$ 0.29	1.49 / 6.75
FedRep+CFE	89.50 $\pm$ 0.06 / 61.23 $\pm$ 0.05	0.85 / 1.81

As shown in Table V, on the datasets CIFAR-10 and CIFAR-100, after integrating other FL algorithms with the FedCFE algorithm, the performance of all algorithms have been enhanced to varying degrees when $\alpha = 0.1$.

E. Scalability experiment of FedCFE

As shown in Table VI, when the number of clients grows from 20 to 100 and 500, performance degradation is consistently observed across all examined algorithms due to increasingly severe data heterogeneity and higher communication variance. Classical FL approaches (FedAvg, FedProx) deteriorate sharply, suggesting poor scalability under large-scale heterogeneous environments. Personalized FL methods (Per-FedAvg, APFL, Ditto) demonstrate better adaptability, yet still suffer notable accuracy loss as the client count reaches 500.

The FedCFE consistently yields the best performance under all client scales, achieving 59.51%, 49.01%, and 36.14% accuracy at 20, 100, and 500 clients, respectively. The results validate that the FedCFE maintains robustness and personalization capability even under large-scale client populations, supporting its viability for large-scale IoT networks.

F. Efficiency and System Cost Analysis

Table VII presents a comparative analysis between the proposed FedCFE method and several representative FL

TABLE VI: Scalability performance of the FedCFE for large-scale IoT clients on CIFAR-100 (Dirichlet($\alpha = 0.1$))

Method	Accuracy (%)		
	Client=20	Client=100	Client=500
FedAvg [12]	30.49 $\pm$ 0.45	23.97 $\pm$ 0.18	15.38 $\pm$ 0.20
Per-FedAvg [15]	36.87 $\pm$ 0.90	19.87 $\pm$ 0.11	21.74 $\pm$ 0.69
APFL [16]	57.35 $\pm$ 0.23	45.41 $\pm$ 0.11	36.01 $\pm$ 0.12
FedProx [18]	30.66 $\pm$ 0.04	21.98 $\pm$ 0.52	9.13 $\pm$ 0.03
Ditto [17]	54.33 $\pm$ 0.20	40.95 $\pm$ 0.12	32.33 $\pm$ 0.04
FedALA [23]	57.39 $\pm$ 0.53	42.12 $\pm$ 0.61	34.00 $\pm$ 0.23
FedCFE (Ours)	59.58 $\pm$ 0.30	49.01 $\pm$ 0.05	36.14 $\pm$ 0.12

algorithms across five dimensions: model accuracy, computational cost, communication cost, memory usage, and prediction model parameter size. All metrics are normalized by setting FedAvg=1.0 and expressed as relative ratios. The experiments are conducted on the CIFAR-100 dataset under a heterogeneous setting with Dirichlet($\alpha = 0.1$). The FedCFE adopts a client participation rate of 0.5 per training round. In addition, for clients not participating in the current training round, the FedCFE still distributes the shared feature extractor module from the aggregated global model, allowing these clients to update their local shared feature space.

As shown in Table VII, the proposed FedCFE algorithm achieves the highest performance in terms of Accuracy, improving by approximately 95% compared with the baseline FedAvg and slightly outperforming strong competitors, such as FedALA and APFL. This result demonstrates the effectiveness of the proposed combination of the CFE, knowledge distillation mechanism, and CG, which jointly enhance the generalization and personalization capabilities of local models. In terms of the computation overhead column, the FedCFE requires about 2.36 times the average computation time per round compared with the FedAvg. This increase mainly arises from the need for each client not only to train the predictive model but

TABLE VII: Performance comparison of PFL algorithms relative to the FedAvg in terms of accuracy, computation overhead, communication overhead, etc. (expressed as multiples)

Method	Accuracy	Computation overhead	Communication overhead	Memory	Prediction model parameters
FedAvg [12]	$\times 1.0$	$\times 1.0$	$\times 1.0$	$\times 1.0$	$\times 1.0$
Per-FedAvg [15]	$\times 1.21$	$\times 1.28$	$\times 1.0$	$\times 1.5$	$\times 1.0$
APFL [16]	$\times 1.88$	$\times 2.30$	$\times 1.0$	$\times 1.5$	$\times 1.0$
FedProx [18]	$\times 1.01$	$\times 1.34$	$\times 1.0$	$\times 1.5$	$\times 1.0$
Ditto [17]	$\times 1.78$	$\times 1.52$	$\times 1.0$	$\times 1.5$	$\times 1.0$
FedALA [23]	$\times 1.88$	$\times 1.19$	$\times 1.0$	$\times 1.55$	$\times 1.0$
FedCFE (Ours)	$\times 1.95$	$\times 2.36$	$\times 0.75$	$\times 1.7$	$\times 0.76$

also to update the CG and perform cross-client knowledge distillation, thereby increasing computational complexity. Nevertheless, the computation cost is comparable to that of APFL ($\times 2.3$) and remains within an acceptable range, indicating that the FedCFE achieves a favorable trade-off between performance improvement and computational efficiency. In terms of the communication overhead column, the FedCFE requires only 0.75 times the communication volume of the FedAvg, which is significantly lower than that of other algorithms. This advantage stems from the fact that the FedCFE only requires clients to upload the fine-tuned last-layer parameters of the CG, a small number of soft labels, and scaled representations of local data distributions, while the server only needs to transmit the shared feature extractor module of the global model. This design reduces the transmission of full model gradients and intermediate features, thus achieving effective communication compression. In terms of the prediction model parameters column, the FedCFE uses only 0.76 times the number of parameters of the FedAvg, showing that the proposed design achieves significant advantages in model compression and feature extraction. Through the modular combination mechanism, the FedCFE attains equal or even superior accuracy compared with other advanced personalized FL algorithms while maintaining fewer model parameters.

In summary, the FedCFE enhances model accuracy while maintaining acceptable computational and memory costs, and it exhibits clear advantages in communication efficiency and parameter compactness. These results suggest that the proposed method offers strong practicality and system-level balance for FL in edge computing and resource-constrained environments.

V. CONCLUSION

We proposed a novel PFL algorithm, FedCFE, which combines the personalized feature extractor with the public feature extractor in parallel to construct a highly adaptable personalized model. By treating the outputs of both extractors as the input of a personalized classifier, the FedCFE fully exploits and integrates shared and client-specific representations, thereby improving model performance under heterogeneous federated learning scenarios. Experimental results demonstrate that the FedCFE

achieves superior robustness and effectiveness, particularly on datasets with high statistical heterogeneity. Meanwhile, FedCFE achieves a reasonable trade-off between accuracy and efficiency in terms of computational overhead, communication cost, memory usage, and model size, reflecting good lightweight deployability. Furthermore, performance gains remain consistent across both visual and non-visual IoT datasets, validating the versatility and strong generalization capability of the FedCFE.

While this work primarily focuses on algorithmic design and simulation-based evaluation, due to limited hardware resources, we have not yet performed real-device energy measurements or end-to-end deployment studies. Future work will therefore investigate large-scale and heterogeneous IoT deployments. It will also evaluate the FedCFE on real hardware platforms in terms of energy consumption, latency, and scalability, integrate privacy-enhancing techniques, thereby strengthening the practicality of the FedCFE in complex IoT ecosystems.

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APPENDIX

TABLE A1: Symbols during the server pre-training

Symbol	Description
$w_{c,s}$	PreCG model on server
w_d	Discriminator
$\mathbf{E}$	Embedded vector composed of label vector and noise vector
$\mathcal{L}_{CG,s}$	Loss function of server-side PreCG

TABLE A2: Symbols during the server-side personalized feature extractor group-weighted aggregation mechanism

Symbol	Description
$\bar{w}_{p,k}^{(t)}$	Personalized feature extractor of client k during the t -th round of communication
$\bar{w}_p^{(t,j)}$	Aggregated personalized feature extractor corresponding to the j -th client cluster during the t -th round of communication
$\mathcal{C}_j$	The j -th client cluster

TABLE A3: Symbols during the client-side CFE training

Symbol	Description
L	Average loss function of a batch or of all samples
ℓ	Loss function of a single data sample
$w_{CFE,k}$	Composite feature extractor of client k
$w_{CFE}^{(e)}$	CFE of the e -th local iteration
$\bar{w}_s : \mathbb{R}^h \rightarrow \mathbb{R}^a$	Shared feature extractor maps the high-dimensional local data of the client into the low-dimensional common features
$\bar{w}_p : \mathbb{R}^h \rightarrow \mathbb{R}^b$	Personalized feature extractor maps the high-dimensional local data of the client into the low-dimensional personalized features
$\bar{w}_{p,k}$	Personalized feature extractor of client k
$\bar{w}_{p,k}^{(e)}$	CFE of the e -th local iteration regarding personalized feature extractor of client k
$w_{p,k}$	Personalized model of client k
$h_{p,k}$	Personalized classifier of client k
$h_p^{(e)}$	CFE of the e -th local iteration regarding personalized classifier

TABLE A4: Symbols during the local CG model fine-tuning

Symbol	Description
$D_k^{(y)}$	Label set of client k
$D_{c,k}^{(x)}$	Feature vectors generated by local CG of client k
ε	Interpolation weight of loss function of client CG
$L_{CE,c}$	CE loss function of local CG
$L_{KL,c}$	KL loss function in local fine-tuning CG
$L_{CG,k}$	Total loss function of local CG of client k
$w_{c,k}^{(t,e)}$	CG of client k during the e -th iteration of the t -th round
$w_{CFE,k}^{(t,e)}$	CFE of client k during the e -th iteration of the t -th round
$w_{c,k,h}^{(t,e)}$	The last h layers of the CG of the client k during the e -th iteration of the t -th round
$w_{c,l}$	The remaining low-dimensional l layers in the CG
m	Proportion of label data for each client k is extracted

TABLE A5: Symbols during the server-side knowledge distillation

Symbol	Description
$D_{k,m}^{(x)}$	$m\%$ feature vectors generated by the CG of client k
$D_{k,m}^{(y)}$	A subset of $m\%$ of the labeled data sampled from dataset D_k of client k
$w_{c,k}^{(t+1)}$	CG uploaded by client k during the t -th round of communication
$w_s^{(t)}$	Server model during the t -th round of communication
$L_{KL,s,k}$	KL loss function between the soft labels uploaded by client k and the soft labels of the data generated by $w_{c,k}^{(t+1)}$ is calculated
$L_{CE,s,k}$	CE loss function of the data generated by the client k
L_{KD}	Total loss function of knowledge distillation
$P'_{c,k}(x_{c,k})$	Soft label calculated from the generated dataset
$P'_{k,m}$	Soft labels corresponding to $m\%$ samples in dataset of client k
β	Interpolation weight of distillation loss function of server student model
η	Learning rate of model